Lab - Data Visualization using Excel

Pre-Task Guidelines:

- Before beginning the tasks, please adhere to these guidelines.
- Create a workbook titled BI_Analytics_Capstone_ML (follow the format of M03L01) and save it.
- Import all the Excel files listed below into this workbook, allocating each to its own worksheet.
- Ensure that you name each worksheet in a way that corresponds with the file names.

Data:

Use the provided files - *sales.xlsx*, *product_hierarchy.xlsx*, *store_cities.xlsx*, *store_names.xlsx*, *city_names.xlsx*, and *product_names.xlsx.*

Hold the CTRL key while you click each file to download it:

sales.xlsx

product_hierarchy.xlsx

store_cities.xlsx

store_names.xlsx

city_names.xlsx

product_names.xlsx

Software: MS Excel

Duration: 1.5 hr

Tasks:

To achieve a comprehensive understanding and visualization of sales data, you'll perform the following tasks using advanced Excel features.

1. Data Compilation and Integration

- Use the six provided Excel files: sales_.xlsx (cleaned sales data), product_hierarchy.xlsx (product categories), product_names.xlsx (names of products), store_cities.xlsx (cities where stores are located), store_names.xlsx (names of stores), and city_names.xlsx (names of cities), which you have already imported in a workbook as a prerequisite of this lesson.
- Use Excel's Power Query or VLOOKUP to combine these files into a single, consolidated workbook. This requires carefully aligning related columns (like product_id, store_id, and city_id) across the different datasets to ensure accurate consolidation.
- Take a screenshot.

2. Basic Chart Creation

In this task, you will visualize fundamental sales information through basic charting techniques.

- Create a bar chart with the consolidated data to display the total sales per store. This will help you quickly identify which stores are outperforming or underperforming in sales.
- Develop a line chart to visualize how sales trends evolve over a specified timeframe. This is crucial for identifying seasonal trends, growth, or decline in sales.
- Take a screenshot.

3. Create Advanced Charts for Hierarchical Data Visualization

In this task, you will dive deeper into the sales data with more complex visualizations to uncover nuanced insights.

- Prepare a separate worksheet from the combined Excel file, taking the fields city_id, store_id, and product_id and sales data. This preparation is crucial for structuring the data in a hierarchical manner suitable for a Sunburst Chart, which will clearly show the sales distribution of each product of every store of each city.
- Take a screenshot.

▶ **Hint:** Your Sunburst Chart should look like the diagram below.

4. Analyze Relationships with Scatter Plots

- Create a scatter plot to explore the relationship between sales volumes and stock levels. This analysis is key to understanding how stock availability might affect sales.
- Enhance the scatter plot by adding a trendline. This will clearly indicate the general direction or correlation between sales and stock, offering insights into whether there's a positive, negative, or no correlation.
- Take a screenshot.

5. Take Screenshots

- Ensure you have taken a screenshot after each task, as you'll be required to add them to your final presentation.

Note

By completing these tasks, you'll demonstrate that you have mastered the technical skills needed to manipulate and visualize data in Excel and gained the ability to extract meaningful insights from complex datasets, a crucial skill in data-driven decision-making processes.

Make sure to capture screenshots for each task, as you'll be required to add them to your final PowerPoint presentation.

Congratulations! You have completed this lab and are ready for the next topic.

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